Last-Day Revision Review: Linear System Theory

Scope: short final-review document for the proof-heavy part of the course. It is not a full replacement for the audited lecture notes. It prioritizes definitions, theorem statements, notation traps, and complete proof formulations that are likely to be asked.

1. Professor's Likely Exam Pattern

The most likely questions are not long numerical computations. Expect short theoretical questions of the following forms:

- State the definition and prove the immediate consequence.
- Show why a special matrix structure implies a spectral or geometric property.
- Distinguish two similar-looking notions, especially over real versus complex fields.
- Prove a projection, unitary, Hermitian, normal, PSD, or SVD property.
- State a theorem precisely and give the proof idea.
- Explain what a factorization reveals and what hypotheses it needs.

The final is inclusive, but later lectures are more likely to dominate: Schur, normal matrices, spectral theorem, PSD/PD, Cholesky, QR, Schur complements, matrix norms, SVD, low-rank approximation, polar decomposition, and nuclear norm.

2. Definitions to Memorize Exactly

Inner Product and Norm

For complex vectors, the standard inner product is

$$\langle x, y\rangle = y^* x$$

or equivalently $x^* y$, depending on the course convention. Be consistent with the convention used in the proof. The induced norm is

$$\|x\| = \sqrt{x^* x}.$$

Vectors are orthogonal if

$$x^* y = 0.$$

Unitary and Orthogonal Matrices

A complex square matrix U is unitary if

$$U^* U = U U^* = I.$$

A real square matrix Q is orthogonal if

$$Q^T Q = Q Q^T = I.$$

Projection and Orthogonal Projection

A matrix P is a projection if

$$P^2 = P.$$

It is an orthogonal projection if additionally

$$P^* = P.$$

If Q has orthonormal columns spanning a subspace V, then the orthogonal projection onto V is

$$P = QQ^*.$$

If A has full column rank and $\mathcal{R}(A) = V$, then

$$P = A(A^*A)^{-1}A^*.$$

Hermitian, Symmetric, Skew-Hermitian

A complex matrix is Hermitian if

$$A^* = A.$$

A real matrix is symmetric if

$$A^T = A.$$

A complex matrix is skew-Hermitian if

$$A^* = -A.$$

Positive Definite and Positive Semidefinite

A Hermitian matrix A is positive definite if

$$x^*Ax > 0 \quad \text{for all } x \neq 0.$$

It is positive semidefinite if

$$x^*Ax \geq 0 \quad \text{for all } x.$$

Write $A \succ 0$ for positive definite and $A \succeq 0$ for positive semidefinite.

Normal Matrix

A square complex matrix A is normal if

$$A^*A = AA^*.$$

Hermitian, skew-Hermitian, and unitary matrices are normal, but normal matrices need not be Hermitian.

Matrix Norms

The induced norm from vector norms is

$$\|A\|_{p,p} = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}.$$

Important cases:

$$\|A\|_{2,2} = \sigma_{\max}(A),$$

$$\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2} = \sqrt{\sum_i \sigma_i^2},$$

$$\|A\|_* = \sum_i \sigma_i.$$

3. Theorems to State Exactly

Schur Theorem

Every square complex matrix $A \in \mathbb{C}^{n \times n}$ admits a factorization

$$A = UTU^*,$$

where U is unitary and T is upper triangular. The diagonal entries of T are the eigenvalues of A.

Spectral Theorem for Hermitian Matrices

If $A = A^*$, then

$$A = U\Lambda U^*,$$

where U is unitary and Λ is real diagonal.

For real symmetric A,

$$A = Q\Lambda Q^T,$$

with Q orthogonal and Λ real diagonal.

Normal Matrix Theorem

A square complex matrix A is normal if and only if it is unitarily diagonalizable:

$$A^*A = AA^* \quad \Longleftrightarrow \quad A = U\Lambda U^*.$$

SVD Theorem

Every $A \in \mathbb{C}^{m \times n}$ has a singular value decomposition

$$A = U\Sigma V^*,$$

where U and V are unitary and Σ is rectangular diagonal with nonnegative singular values.

Cholesky Theorem

If $A = A^* \succ 0$, then

$$A = LL^*,$$

where L is lower triangular with positive diagonal entries. In the real symmetric case, $A = LL^T$.

Schur Complement PD Test

For a Hermitian block matrix

$$M = \begin{bmatrix} A & B \\ B^* & C \end{bmatrix},$$

with $A \succ 0$,

$$M \succ 0 \quad \Longleftrightarrow \quad A \succ 0 \text{ and } C - B^* A^{-1} B \succ 0.$$

4. Complete Proofs Most Likely to Be Asked

Proof 1: Unitary Matrices Preserve Inner Products and Norms

Claim: If $U^*U = I$, then

$$\langle Ux, Uy \rangle = \langle x, y \rangle$$

and

$$\|Ux\| = \|x\|.$$

Proof:

$$\langle Ux, Uy \rangle = (Uy)^*(Ux) = y^* U^* U x = y^* x = \langle x, y \rangle.$$

Taking $y = x$,

$$\|Ux\|^2 = (Ux)^*(Ux) = x^* U^* U x = x^* x = \|x\|^2.$$

Since norms are nonnegative,

$$\|Ux\| = \|x\|.$$

Likely exam wording: “Show that unitary matrices preserve lengths and angles.”

Common mistake: using U^T instead of U^* in the complex case.

Proof 2: $P = QQ^*$ Is an Orthogonal Projection

Assume $Q \in \mathbb{C}^{m\times k}$ has orthonormal columns, so

$$Q^*Q = I_k.$$

Let

$$P = QQ^*.$$

Then

$$P^2 = (QQ^*)(QQ^*) = Q(Q^*Q)Q^* = QIQ^* = QQ^* = P.$$

Also,

$$P^* = (QQ^*)^* = (Q^*)^*Q^* = QQ^* = P.$$

Therefore P is an orthogonal projection.

Likely exam wording: "Prove that QQ^* is the orthogonal projection onto the column space of Q."

To finish the range statement:

- If $y = Qc \in \mathcal{R}(Q)$, then $Py = QQ^*Qc = Qc = y$.
- If $z \perp \mathcal{R}(Q)$, then $Q^*z = 0$, so $Pz = 0$.

Common mistake: claiming $QQ^* = I$ when Q is tall. Only $Q^*Q = I$ is guaranteed.

Proof 3: Projection Formula $P = A(A^*A)^{-1}A^*$

Assume $A \in \mathbb{C}^{m\times n}$ has full column rank. Then A^*A is invertible.

Define

$$P = A(A^*A)^{-1}A^*.$$

Idempotence:

$$P^2 = A(A^*A)^{-1}A^*A(A^*A)^{-1}A^* = A(A^*A)^{-1}(A^*A)(A^*A)^{-1}A^* = P.$$

Hermitian property:

$$P^* = \left[A(A^*A)^{-1}A^*\right]^* = A\left[(A^*A)^{-1}\right]^* A^*.$$

Since A^*A is Hermitian positive definite, its inverse is Hermitian, so

$$\left[(A^*A)^{-1}\right]^* = (A^*A)^{-1}.$$

Thus

$$P^* = P.$$

Therefore P is an orthogonal projection onto $\mathcal{R}(A)$.

Likely exam wording: "Derive or verify the projection matrix onto $\mathcal{R}(A)$ for full column rank A."

Common mistake: forgetting the full-column-rank condition needed for $(A^*A)^{-1}$.

Proof 4: Normal Equations for Least Squares

Problem:

$$\min_x \|Ax - b\|_2^2.$$

Let

$$f(x) = (Ax - b)^*(Ax - b).$$

In the real case, expand:

$$f(x) = x^T A^T Ax - 2x^T A^T b + b^T b.$$

Setting the gradient to zero gives

$$A^T Ax = A^T b.$$

In the complex case, the normal equations are

$$A^* Ax = A^* b.$$

Geometric proof:

At the minimizer $\hat{x}$, the residual

$$r = b - A\hat{x}$$

is orthogonal to the column space of A. Therefore

$$A^* r = 0,$$

so

$$A^*(b - A\hat{x}) = 0,$$

which gives

$$A^* A\hat{x} = A^* b.$$

Likely exam wording: "Explain the geometric meaning of the normal equations."

Common mistake: treating $A^T A$ as correct in complex least squares.

Proof 5: Hermitian Eigenvalues Are Real

Assume

$$A = A^*$$

and

$$Av = \lambda v, \qquad v \neq 0.$$

Then

$$v^* A v = \lambda v^* v.$$

Since $A = A^*$,

$$(v^* A v)^* = v^* A^* v = v^* A v.$$

So $v^* A v$ is real. Also $v^* v > 0$ is real. Hence

$$\lambda = \frac{v^* A v}{v^* v} \in \mathbb{R}.$$

Likely exam wording: "Prove eigenvalues of Hermitian matrices are real."

Common mistake: writing $v^T A v$ in the complex case.

Proof 6: Hermitian Eigenvectors for Distinct Eigenvalues Are Orthogonal

Assume $A = A^*$,

$$Ax = \lambda x, \qquad Ay = \mu y,$$

with $\lambda \neq \mu$. Since Hermitian eigenvalues are real, $\overline{\mu} = \mu$.

Compute:

$$y^* A x = y^* (\lambda x) = \lambda y^* x.$$

Also, from $Ay = \mu y$,

$$(Ay)^* = y^* A^* = \overline{\mu} y^* = \mu y^*.$$

Since $A^* = A$,

$$y^* A = \mu y^*.$$

Therefore

$$y^* A x = \mu y^* x.$$

Combining,

$$\lambda y^* x = \mu y^* x.$$

Thus

$$(\lambda - \mu)y^* x = 0.$$

Because $\lambda \neq \mu$,

$$y^* x = 0.$$

So x and y are orthogonal.

Likely exam wording: "Prove eigenspaces of a Hermitian matrix corresponding to distinct eigenvalues are orthogonal."

Common mistake: skipping the conjugate-transpose step from $Ay = \mu y$ to $y^* A = \mu y^*$.

Proof 7: PSD Eigenvalue Criterion

Assume $A = A^*$. By the spectral theorem,

$$A = U \Lambda U^*$$

with U unitary and $\Lambda = \operatorname{diag}(\lambda_i)$ real.

For any x, let

$$y = U^* x.$$

Then

$$x^* A x = x^* U \Lambda U^* x = y^* \Lambda y = \sum_i \lambda_i |y_i|^2.$$

If all $\lambda_i \geq 0$, then $x^* A x \geq 0$, so $A \succeq 0$.

Conversely, if $A \succeq 0$, choose $x = u_i$, the i-th eigenvector. Then

$$u_i^* A u_i = \lambda_i u_i^* u_i = \lambda_i \geq 0.$$

Therefore all eigenvalues are nonnegative.

For positive definite, replace ≥ 0 by > 0 for all nonzero x. Then all eigenvalues must be strictly positive.

Likely exam wording: "Show that a Hermitian matrix is PSD iff all eigenvalues are nonnegative."

Common mistake: applying eigenvalue sign criteria to non-Hermitian matrices without qualification.

Proof 8: Gram Matrices Are PSD

Let

$$G = A^* A.$$

Then G is Hermitian:

$$G^* = (A^* A)^* = A^* A = G.$$

For any x,

$$x^*Gx = x^*A^*Ax = (Ax)^*(Ax) = \|Ax\|^2 \geq 0.$$

Therefore

$$A^*A \succeq 0.$$

If A has full column rank, then $Ax \neq 0$ for all $x \neq 0$, so

$$x^*A^*Ax = \|Ax\|^2 > 0,$$

and $A^*A \succ 0$.

Likely exam wording: “Prove A^*A is positive semidefinite and state when it is positive definite.”

Common mistake: saying A^*A is always invertible. It is invertible only when A has full column rank.

Proof 9: Schur Theorem

Claim: For every square complex matrix A, there exists a unitary U and upper triangular T such that

$$A = UTU^*.$$

Proof idea:

Every complex square matrix has at least one eigenvalue λ_1 and eigenvector u_1. Normalize u_1, then complete it to an orthonormal basis:

$$U_1 = [u_1 \ u_2 \ \cdots \ u_n].$$

Then

$$U_1^*AU_1 = \begin{bmatrix} \lambda_1 & * \\ 0 & A_2 \end{bmatrix}.$$

The lower-left block is zero because

$$Au_1 = \lambda_1 u_1.$$

Now apply the same argument recursively to A_2. By induction, the lower-right block can be unitarily triangularized. Combining the unitary transformations gives

$$U^*AU = T,$$

where T is upper triangular. Thus

$$A = UTU^*.$$

Likely exam wording: “State Schur theorem and explain its proof.”

Common mistake: saying Schur gives diagonal form. It gives triangular form.

Proof 10: Normal Implies Unitary Diagonalizable

Assume A is normal:

$$A^*A = AA^*.$$

By Schur theorem,

$$A = UTU^*$$

with T upper triangular. Normality is preserved by unitary similarity, so T is normal.

Now use the fact: an upper triangular normal matrix is diagonal.

Reason: for an upper triangular normal T, compare row and column norm relations from $T^*T = TT^*$. The first column has only t_{11}, while the first row has $t_{11}, t_{12}, \dots, t_{1n}$. Equality of the relevant diagonal entries forces

$$t_{12} = \cdots = t_{1n} = 0.$$

Repeat inductively for the remaining block. Hence all off-diagonal entries are zero.

Thus $T = \Lambda$, and

$$A = U\Lambda U^*.$$

Likely exam wording: "Use Schur theorem to prove that normal matrices are unitarily diagonalizable."

Common mistake: trying to prove this by assuming eigenvectors are already orthogonal.

Proof 11: SVD Existence

Start with

$$A^*A.$$

It is Hermitian PSD, so by the spectral theorem,

$$A^*A = V\Lambda V^*,$$

where V is unitary and

$$\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n), \qquad \lambda_i \geq 0.$$

Define singular values

$$\sigma_i = \sqrt{\lambda_i}.$$

For each positive σ_i, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

Then

$$Av_i = \sigma_i u_i.$$

Check orthonormality:

$$u_i^* u_j = \frac{v_i^* A^* A v_j}{\sigma_i \sigma_j} = \frac{v_i^* \lambda_j v_j}{\sigma_i \sigma_j}.$$

If $i \neq j$, then $v_i^* v_j = 0$, so $u_i^* u_j = 0$. If $i = j$, then

$$u_i^* u_i = \frac{\lambda_i}{\sigma_i^2} = 1.$$

Complete the u_i's to an orthonormal basis of the output space. With U and V built from these bases,

$$A = U \Sigma V^*.$$

Likely exam wording: "Prove or explain why SVD exists."

Common mistake: saying singular values are eigenvalues of A. They are square roots of eigenvalues of $A^* A$.

Proof 12: Best Rank-k Approximation by Truncated SVD

Statement: If

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^*$$

with $\sigma_1 \geq \cdots \geq \sigma_r > 0$, then

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*$$

is a best rank-k approximation to A in the spectral norm and Frobenius norm.

Expected proof idea, not full long proof:

Unitary invariance reduces the problem to approximating Σ:

$$\|A - B\| = \|U^*(A - B)V\| = \|\Sigma - U^* B V\|.$$

Since $U^* B V$ has rank at most k, it cannot match all singular directions beyond the first k. The best choice keeps the largest k singular values and discards the rest.

Consequences:

$$\|A - A_k\|_2 = \sigma_{k+1},$$

$$\|A - A_k\|_F = \sqrt{\sum_{i>k} \sigma_i^2}.$$

Likely exam wording: "State the low-rank approximation theorem and explain why truncating the SVD is optimal."

Common mistake: truncating eigenvalues of A instead of singular values.

Proof 13: Cholesky via Schur Complement

Assume $A = A^* \succ 0$. Partition

$$A = \begin{bmatrix} \alpha & b^* \\ b & C \end{bmatrix}.$$

Since $A \succ 0$,

$$\alpha > 0.$$

Set

$$L = \begin{bmatrix} \sqrt{\alpha} & 0 \\ \frac{b}{\sqrt{\alpha}} & L_2 \end{bmatrix}.$$

Then matching LL^* requires

$$L_2 L_2^* = C - \frac{1}{\alpha} bb^*.$$

The matrix

$$C - \frac{1}{\alpha} bb^*$$

is the Schur complement of α in A, and it is positive definite because $A \succ 0$ and $\alpha > 0$. Therefore, by induction, it has a Cholesky factor L_2. This gives

$$A = LL^*.$$

Likely exam wording: "Explain the proof of Cholesky factorization using Schur complements."

Common mistake: saying subtracting a PSD rank-one term from a PD matrix is always PD. It is true here because it is the Schur complement of a PD block matrix.

Proof 14: Schur Complement Positive Definiteness Test

Let

$$M = \begin{bmatrix} A & B \\ B^* & C \end{bmatrix}, \qquad A \succ 0.$$

Factor:

$$M = \begin{bmatrix} I & 0 \\ B^* A^{-1} & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & C - B^* A^{-1} B \end{bmatrix} \begin{bmatrix} I & A^{-1} B \\ 0 & I \end{bmatrix}.$$

The left and right factors are conjugate transposes of each other:

$$\begin{bmatrix} I & A^{-1} B \\ 0 & I \end{bmatrix}^* = \begin{bmatrix} I & 0 \\ B^* A^{-1} & I \end{bmatrix}.$$

Congruence by an invertible matrix preserves positive definiteness. Therefore,

$$M \succ 0$$

if and only if

$$\begin{bmatrix} A & 0 \\ 0 & C - B^* A^{-1} B \end{bmatrix} \succ 0.$$

This holds if and only if

$$A \succ 0 \quad \text{and} \quad C - B^* A^{-1} B \succ 0.$$

Likely exam wording: “Prove the Schur complement criterion for positive definiteness.”

Common mistake: using ordinary similarity instead of congruence for definiteness.

Proof 15: Matrix 2-Norm Equals Largest Singular Value

Let

$$A = U \Sigma V^*.$$

Then

$$\|Ax\|_2 = \|U \Sigma V^* x\|_2.$$

Unitary matrices preserve norms, so if $y = V^* x$, then $\|y\|_2 = \|x\|_2$, and

$$\frac{\|Ax\|_2}{\|x\|_2} = \frac{\|\Sigma y\|_2}{\|y\|_2}.$$

Now

$$\|\Sigma y\|_2^2 = \sum_i \sigma_i^2 |y_i|^2 \leq \sigma_1^2 \sum_i |y_i|^2 = \sigma_1^2 \|y\|_2^2.$$

Therefore

$$\|A\|_{2,2} \leq \sigma_1.$$

Equality is attained by taking $y = e_1$, equivalently $x = v_1$. Then

$$\|Av_1\|_2 = \sigma_1.$$

Thus

$$\|A\|_{2,2} = \sigma_1.$$

Likely exam wording: “Prove that the induced 2-norm is the largest singular value.”

Common mistake: using largest eigenvalue of A instead of largest singular value.

5. Factorizations Last-Day Table

Factorization	Form	Hypothesis	What it reveals
LU	$A = LU$ or $PA = LU$	pivot assumptions	triangular solves
QR	$A = QR$	general; clean reduced form if full column rank	orthonormal basis for range
Schur	$A = UTU^*$	square complex	triangular form, eigenvalues on diagonal
Spectral	$A = U\Lambda U^*$	Hermitian	real eigenvalues, orthonormal eigenbasis
Normal diagonalization	$A = U\Lambda U^*$	$A^*A = AA^*$	unitary eigenbasis
Cholesky	$A = LL^*$	Hermitian PD	triangular square root
PSD square root	$A = SS^*$	Hermitian PSD	energy/covariance factor
Block LDU	block triangular product	invertible block	Schur complement
Woodbury	$(A + BCD)^{-1}$ formula	compatible inverses	low-rank inverse update
DFT diagonalization	$C = F^*\Lambda F$	circulant	convolution becomes multiplication
SVD	$A = U\Sigma V^*$	any matrix	rank, geometry, norms
Polar	$A = QH$	square nonsingular or partial-isometry generalization	unitary-like times PSD

6. Danger Topics

A^T vs A^*

Use A^T for real transpose. Use A^* for complex conjugate transpose. In complex inner products, projections, Hermitian matrices, unitary matrices, SVD, and PSD definitions, use A^*.

Hermitian vs Symmetric

Hermitian:

$$A^* = A.$$

Symmetric:

$$A^T = A.$$

Symmetric is correct only over real fields.

PSD vs PD

PSD:

$$x^*Ax \geq 0.$$

PD:

$$x^*Ax > 0 \quad x \neq 0.$$

PSD allows zero eigenvalues and may be singular. PD has strictly positive eigenvalues and is invertible.

Projection vs Orthogonal Projection

Projection:

$$P^2 = P.$$

Orthogonal projection:

$$P^2 = P, \qquad P^* = P.$$

Diagonalizable vs Unitarily Diagonalizable

Diagonalizable:

$$A = T\Lambda T^{-1}.$$

Unitarily diagonalizable:

$$A = U\Lambda U^*.$$

Unitary diagonalization is stronger and is equivalent to normality over $\mathbb{C}$.

Schur vs Diagonalization

Schur always gives triangular form for square complex matrices:

$$A = UTU^*.$$

It does not always give diagonal form.

Normal vs Hermitian

Hermitian implies normal, but normal does not imply Hermitian.

Normal:

$$A^*A = AA^*.$$

Hermitian:

$$A^* = A.$$

Singular Values vs Eigenvalues

Singular values are nonnegative and come from A^*A:

$$A^*Av_i = \sigma_i^2 v_i.$$

Eigenvalues of A solve

$$Av = \lambda v.$$

Do not confuse them.

7. Likely Exam Questions and Minimal Answer Skeletons

Question 1

State and prove that an orthogonal projection matrix is Hermitian and idempotent.

Minimal answer:

Use $P = QQ^*$, $Q^*Q = I$. Show $P^2 = P$ and $P^* = P$.

Question 2

Prove eigenvalues of Hermitian matrices are real and eigenvectors for distinct eigenvalues are orthogonal.

Minimal answer:

Use $v^*Av = \lambda v^*v$ for real eigenvalues. For orthogonality, compare $y^*Ax = \lambda y^*x$ and $y^*Ax = \mu y^*x$.

Question 3

State Schur theorem and use it to prove normal matrices are unitarily diagonalizable.

Minimal answer:

Schur gives $A = UTU^*$. Normality transfers to T. Upper triangular normal implies diagonal. Hence $A = U\Lambda U^*$.

Question 4

State the spectral theorem and derive the PSD eigenvalue criterion.

Minimal answer:

Write $A = U\Lambda U^*$, let $y = U^*x$, compute $x^*Ax = \sum_i \lambda_i |y_i|^2$.

Question 5

Prove A^*A is PSD and explain when it is PD.

Minimal answer:

$$x^*A^*Ax = \|Ax\|^2 \geq 0.$$

It is PD iff $Ax \neq 0$ for all $x \neq 0$, i.e. A has full column rank.

Question 6

Prove or explain the existence of SVD.

Minimal answer:

Diagonalize $A^*A = V\Lambda V^*$, set $\sigma_i = \sqrt{\lambda_i}$, define $u_i = Av_i/\sigma_i$, prove orthonormality, complete bases.

Question 7

State and prove the Schur complement positive definiteness criterion.

Minimal answer:

Use block LDU/congruence factorization and show $M \succ 0$ iff $A \succ 0$ and the Schur complement is PD.

Question 8

Prove $\|A\|_2 = \sigma_{\max}(A)$.

Minimal answer:

Use SVD and unitary invariance. Reduce to Σ, bound by σ_1, and attain equality at the first right singular vector.

8. Emergency Last-Night List

Memorize these exact statements:

1. $U^*U = I$ implies inner products and norms are preserved.
2. $P^2 = P$ means projection; $P^* = P$ makes it orthogonal.
3. $P = QQ^*$ and $P = A(A^*A)^{-1}A^*$.
4. $A = A^*$ implies real eigenvalues and orthogonal eigenspaces.
5. $A \succeq 0$ iff $x^*Ax \geq 0$ iff Hermitian eigenvalues are nonnegative.
6. $A^*A \succeq 0$, and $A^*A \succ 0$ iff A has full column rank.
7. Schur: $A = UTU^*$, every square complex matrix.
8. Normal: $A^*A = AA^*$ iff $A = U\Lambda U^*$.
9. SVD: $A = U\Sigma V^*$, every matrix.
10. Cholesky: $A = LL^*$, Hermitian positive definite.
11. Schur complement: $M \succ 0$ iff leading block and Schur complement are PD.
12. $\|A\|_2 = \sigma_{\max}(A)$.

If time is limited, prioritize proofs 1, 2, 5, 6, 7, 9, 10, 11, and 15 from this review.